

Part 3 — Ensembles, Tuning, and Full ML Pipeline

Run dir: /home/antony/masai/capstone/spam_detection/run/part3_20260716_210350_268047

Task 1-2 — Decision Tree: Unconstrained vs. Controlled

	Tree	Train Acc	Test Acc	Gap	Extra
Unconstrained		0.9993	0.7529	0.2464	depth=39, leaves=5038
Controlled (max_depth=5, min_samples_split=20)		0.7566	0.7511	0.0055	test AUC=0.8216

Gap dropped from 0.2464 (unconstrained) to 0.0055 (controlled) — the unconstrained tree memorized training data (a classic high-variance model: it grows greedily to depth 39 / 5038 leaves, fitting noise along with signal, with no mechanism to revisit earlier splits).

Task 3 — Gini vs. Entropy (max_depth=5)

Criterion Test Accuracy

Gini 0.7511

Entropy 0.7490

Task 4 — Random Forest

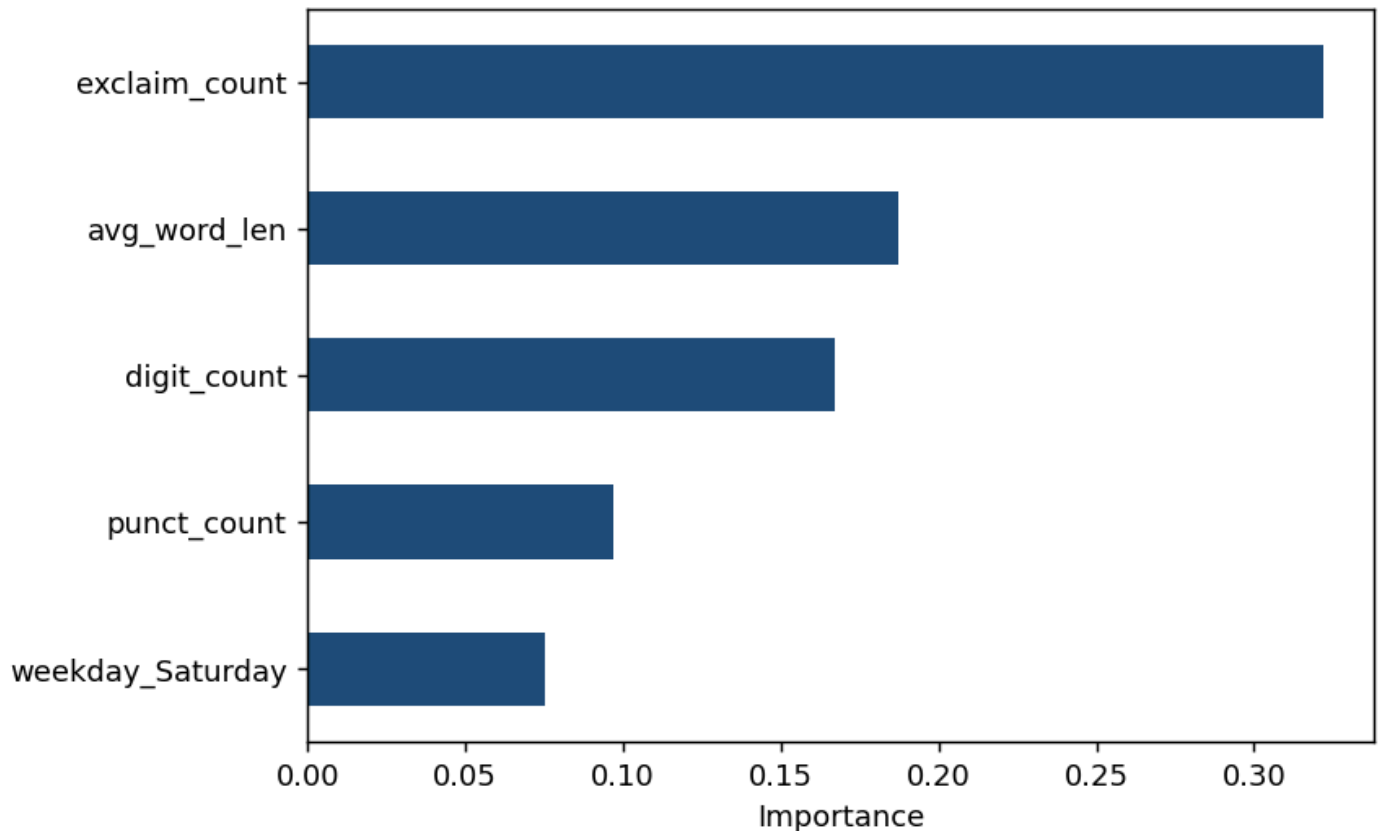
Train Acc Test Acc AUC

0.8172 0.7918 0.8728

Top 5 features by importance

Feature	Importance
exclaim_count	0.3220
avg_word_len	0.1870
digit_count	0.1673
punct_count	0.0969
weekday_Saturday	0.0754

Random Forest - Top 5 Feature Importances



Task 4a — Gradient Boosting

Train Acc	Test Acc	AUC
0.7943	0.7855	0.8655

Task 4b — Feature Ablation

Removed (5 lowest-importance): weekday_Monday, weekday_Thursday, weekday_Wednesday, weekday_Tuesday, capital_ratio

Model	Test AUC
Full (all features)	0.8728
Reduced (5 removed)	0.8789

AUC improved slightly after removal (diff=-0.0060) — these features were **genuinely uninformative**.

Task 5 — Cross-Validated Comparison (5-fold StratifiedKFold, ROC-AUC)

Model	CV Mean AUC	CV Std AUC
Logistic Regression	0.7852	0.0071
Decision Tree (max_depth=5)	0.8244	0.0047
Random Forest	0.8733	0.0037
Gradient Boosting	0.8677	0.0034

Task 6 — GridSearchCV (Random Forest Pipeline)

Configurations evaluated: 18 (x5 folds = 90 fits)

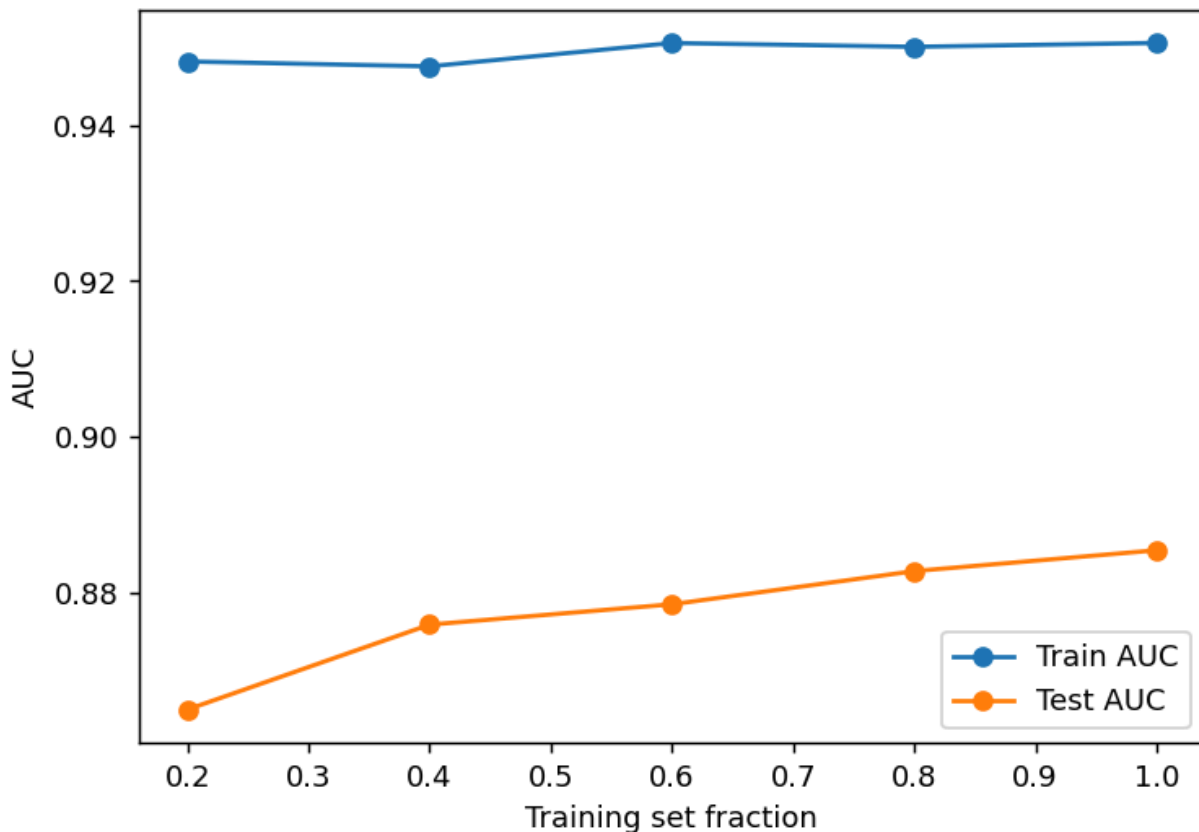
Parameter	Best Value
max_depth	None
min_samples_leaf	5
n_estimators	200

Best CV AUC: **0.8833** | Test AUC: **0.8855**

Task 6b — Learning Curve (Tuned Pipeline)

Training Fraction	N Rows	Train AUC	Test AUC
0.2	4879	0.9481	0.8651
0.4	9758	0.9475	0.8760
0.6	14637	0.9505	0.8786
0.8	19516	0.9500	0.8828
1.0	24395	0.9505	0.8855

Learning Curve - Tuned Random Forest Pipeline



Task 7 — Serialized Model

Saved to: /home/antony/masai/capstone/spam_detection/run/part3_20260716_210350_268047/best_model.pkl

Reload + predict smoke test (2 sample rows): 00

Task 8 — Summary Comparison (Parts 2 + 3)

Model	CV Mean AUC	CV Std AUC	Test AUC
Logistic Regression	0.7852	0.0071	0.7857
Decision Tree (max_depth=5)	0.8244	0.0047	0.8216
Random Forest	0.8733	0.0037	0.8728
Gradient Boosting	0.8677	0.0034	0.8655
Tuned RF Pipeline (GridSearchCV)	0.8833	—	0.8855

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